Personalized Online Course Recommender System

Machine Learning Capstone Project



Christopher Guerrero

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Outline

From exploring the data to two families of recommender systems

1. **Introduction**
   Background, problem statement and approach

2. **Exploratory data analysis**
   Genres, enrollments, popular courses and title keywords

3. **Content-based recommenders**
   Unsupervised: user profiles, course similarity and clustering

4. **Collaborative-filtering recommenders**
   Supervised: KNN, NMF and neural network embeddings

5. **Conclusions and appendix**
   What worked, what to do next, and how to reproduce it

Introduction

Help learners find their next course in a catalog of hundreds

Background and problem

- Learners face **307 courses** across 14 technical genres and need help choosing what to take next.
- A recommender system can **personalize** course suggestions, raising engagement and completions.
- **Problem:** given a learner's enrollment history, which new courses should we recommend?

Approach

- **Content-based (unsupervised):** match courses to each learner's interests using genres, course text and learner clusters.
- **Collaborative filtering (supervised):** learn from all learners' ratings with KNN, NMF and neural network embeddings.
- **Evaluate and compare** both families on held-out data.

33,901
learners

307
courses in the catalog

233,306
enrollments with ratings

95%
of ratings are 3 (completed)

01

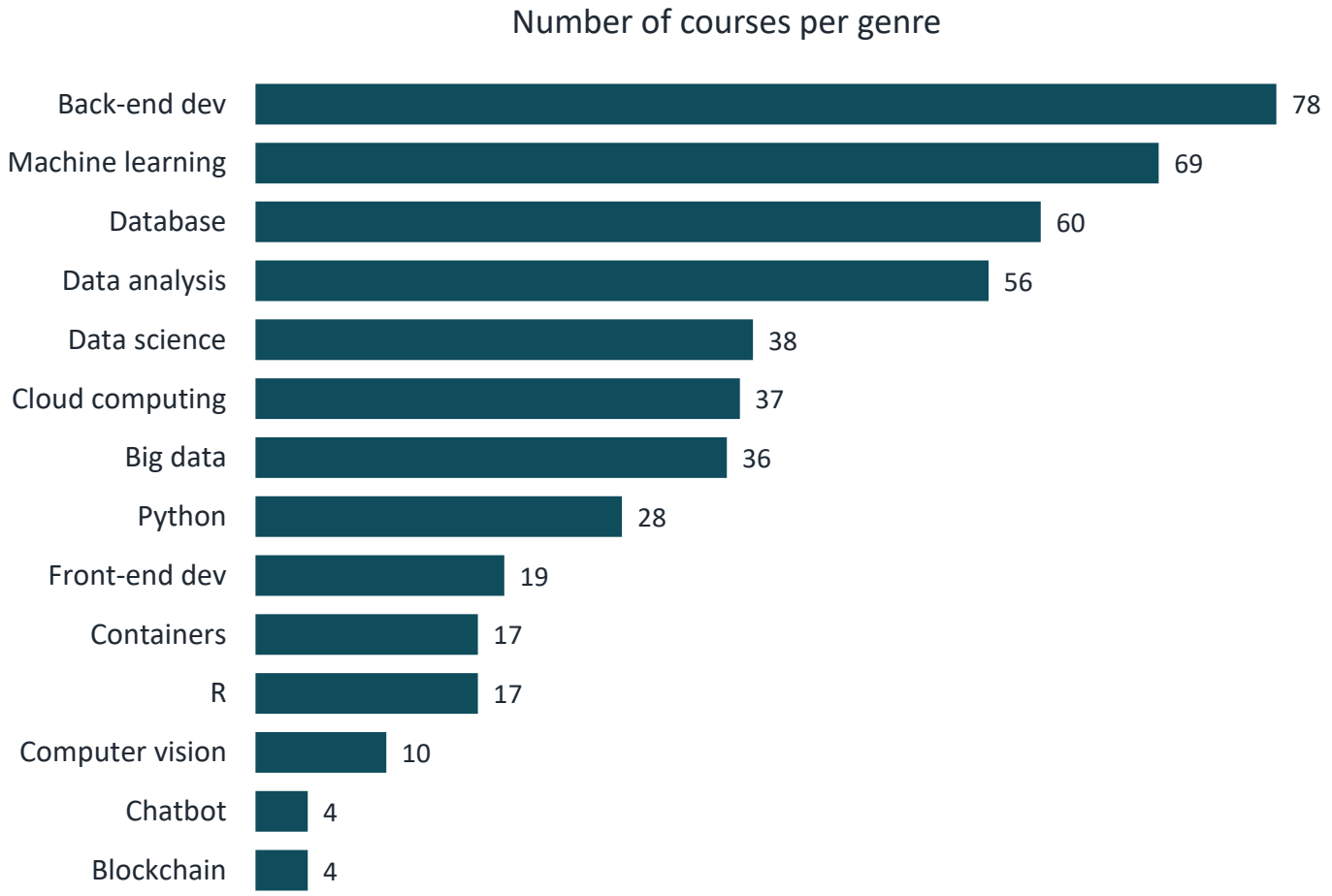
Exploratory Data Analysis



What is in the catalog, and how do learners enroll?

Course counts per genre

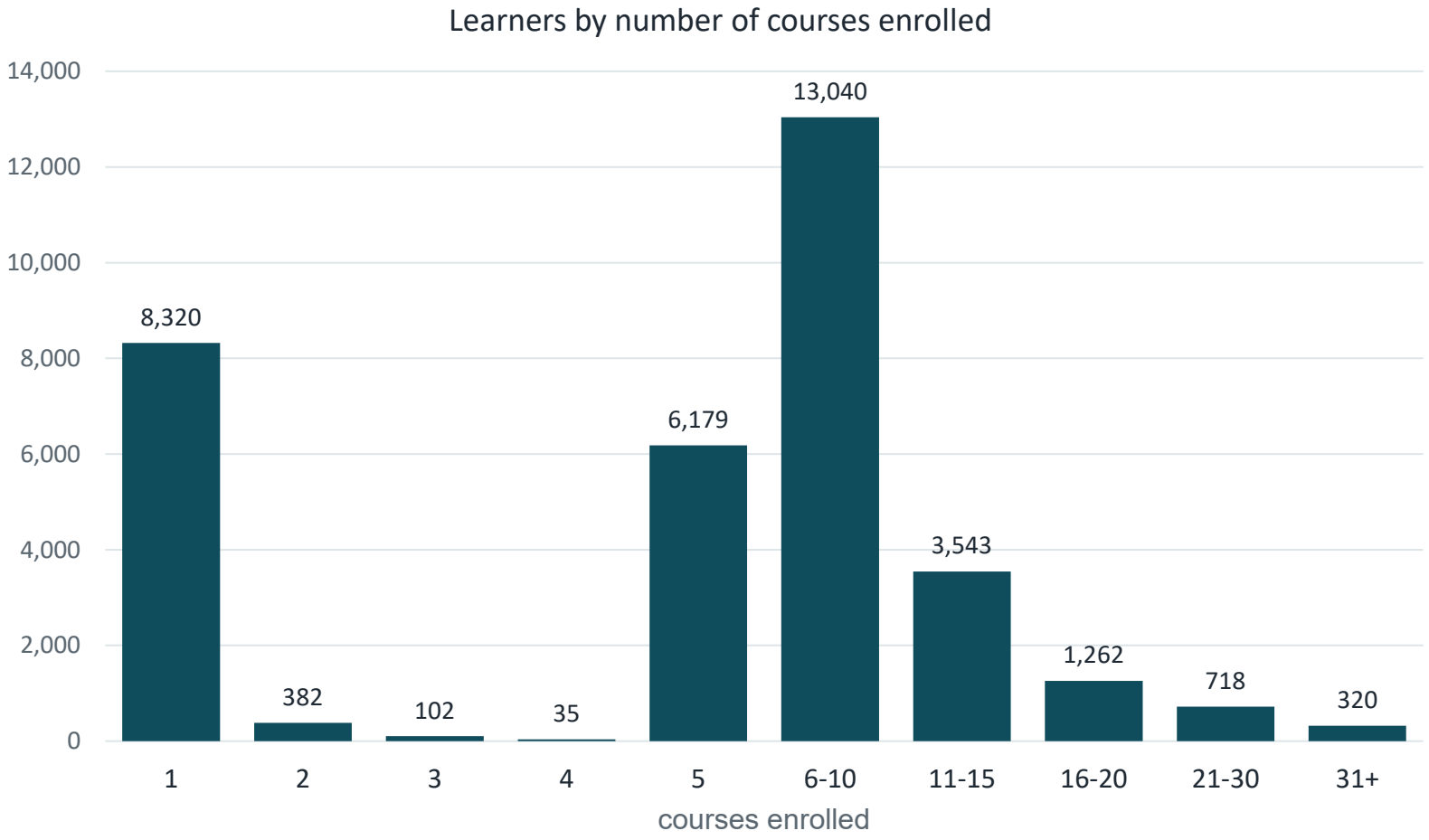
Back-end dev and machine learning lead the catalog; chatbots and blockchain are niche



- **78** back-end development and **69** machine learning courses: the two largest genres.
- Data-related genres (database, data analysis, data science, big data) together cover a large share of the catalog.
- **4 courses each** in chatbots and blockchain: little content to recommend there.
- A course can belong to several genres, so the counts add up to more than 307.

Course enrollment distribution

Most learners take a handful of courses: the median learner enrolled in 6



6

median courses per learner

44%

of learners took 5 or fewer

61

courses: the most active learner

20 most popular courses

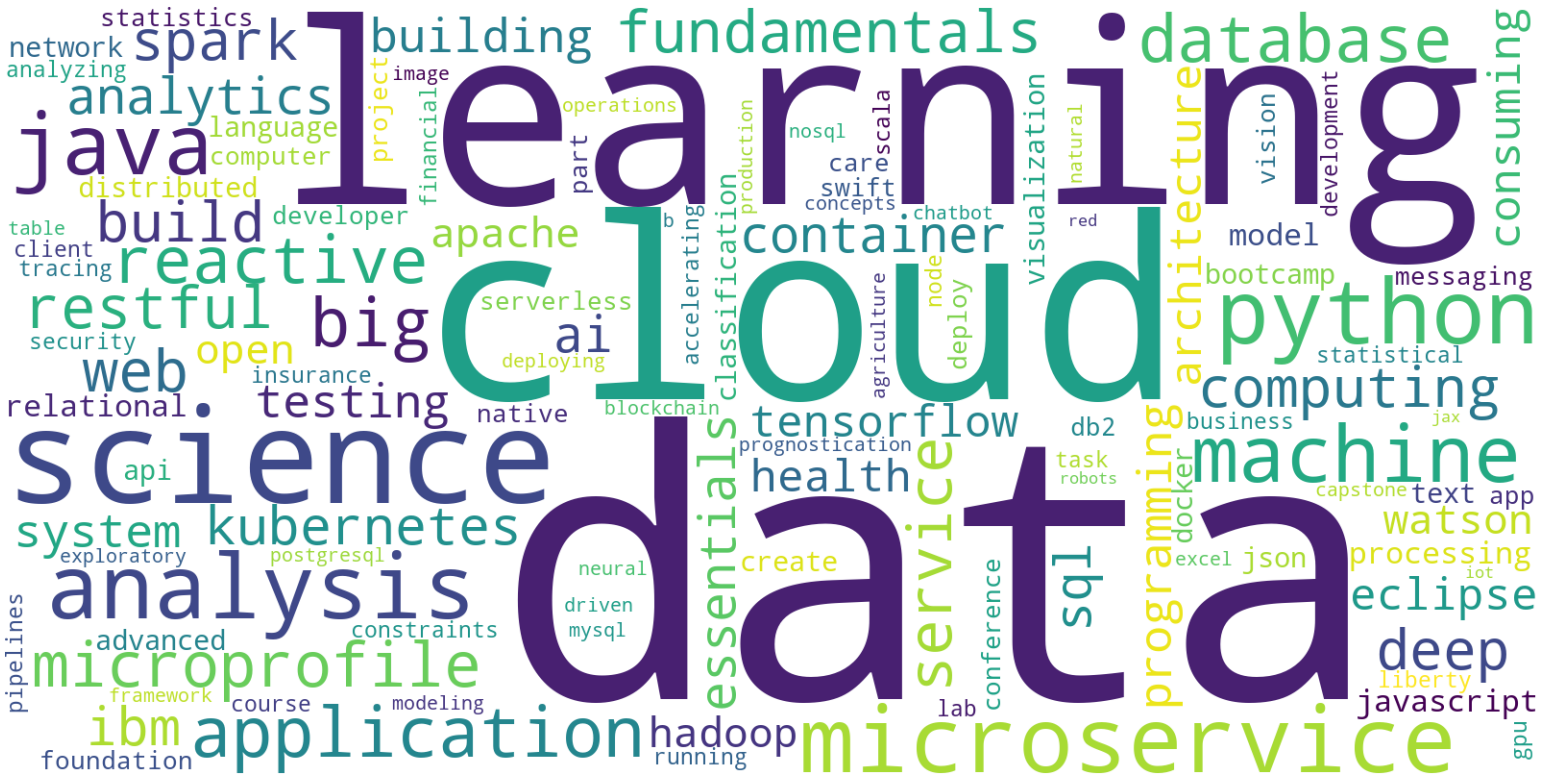
The top 20 of 126 rated courses capture 63% of all enrollments

#	Course	Enrollments
1	Python for Data Science	14,936
2	Introduction to Data Science	14,477
3	Big Data 101	13,291
4	Hadoop 101	10,599
5	Data Analysis with Python	8,303
6	Data Science Methodology	7,719
7	Machine Learning with Python	7,644
8	Spark Fundamentals I	7,551
9	Data Science Hands on with Open Source Tools	7,199
10	Blockchain Essentials	6,719

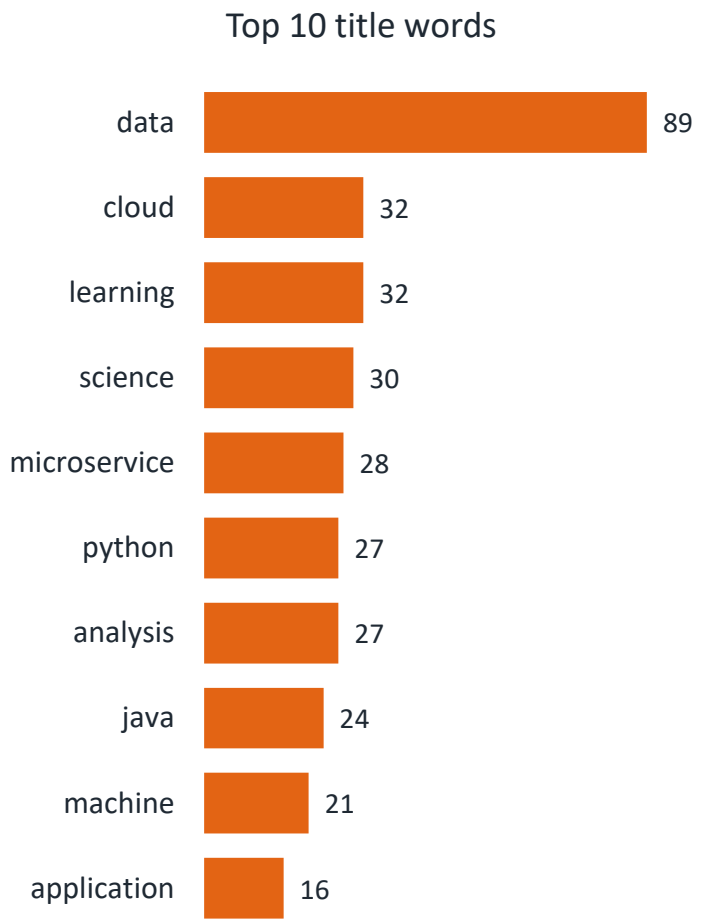
#	Course	Enrollments
11	Data Visualization with Python	6,709
12	Deep Learning 101	6,323
13	Build Your Own Chatbot	5,512
14	R for Data Science	5,237
15	Statistics 101	5,015
16	Introduction to Cloud	4,983
17	Docker Essentials a Developer Introduction	4,480
18	SQL and Relational Databases 101	3,697
19	Mapreduce and Yarn	3,670
20	Data Privacy Fundamentals	3,624

Word cloud of course titles

"data" appears in 89 of 307 titles; cloud, machine learning and Python follow



Word size is proportional to how often the word appears in course titles (common words such as "introduction" and "using" removed).



02

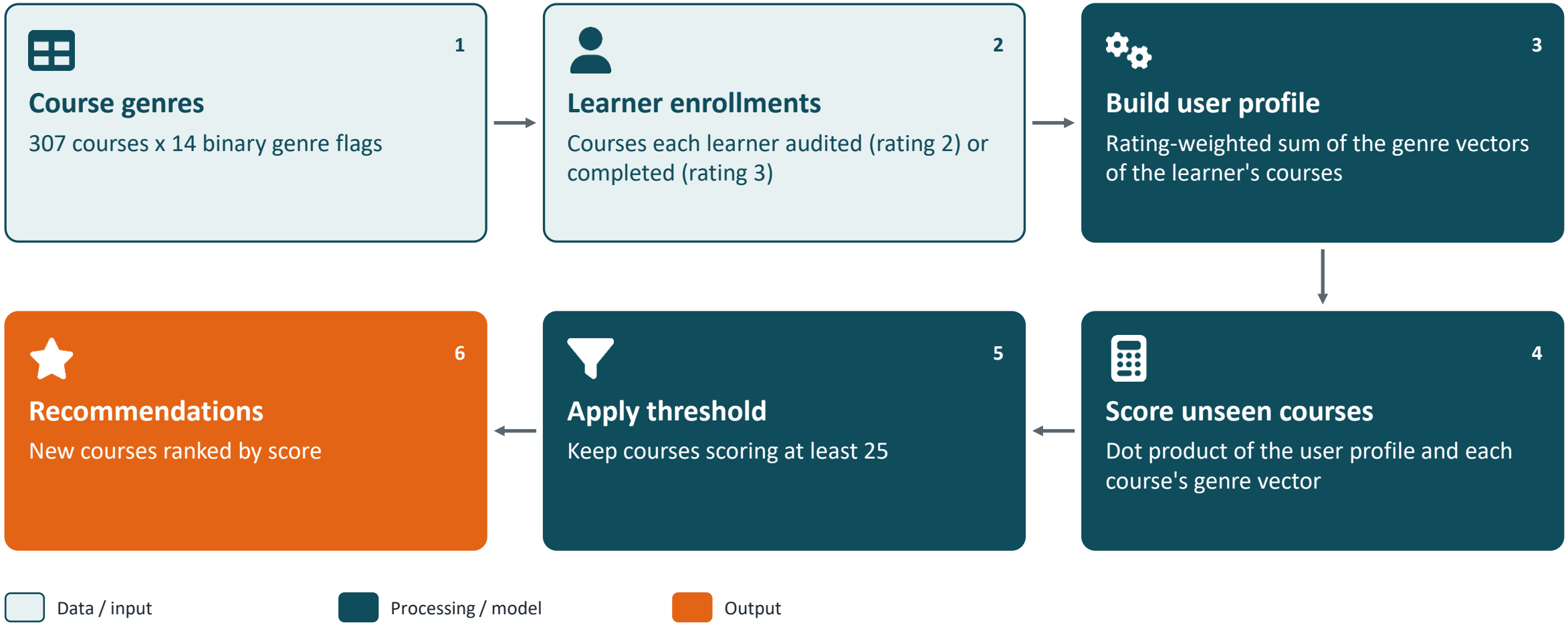
Content-based Recommender Systems

Unsupervised learning on course genres, course text and learner profiles



User profile-based recommender

Flowchart: score unseen courses by how well their genres match a learner's interests



User profile-based: evaluation

Threshold 25 gives 8.2 courses per learner on average, but 60% of learners get none

Hyperparameters

- **Score threshold:** 25, chosen from 10, 15, 20, 25, 30, 35, 40 to give about 10 new courses per learner
- **Score:** dot product of the user profile and the course genre vector
- Heavy learners have large profile values, so a fixed threshold favors them

8.2
new courses per learner (average)

0
new courses (median learner)

60.3%
of learners get no recommendation

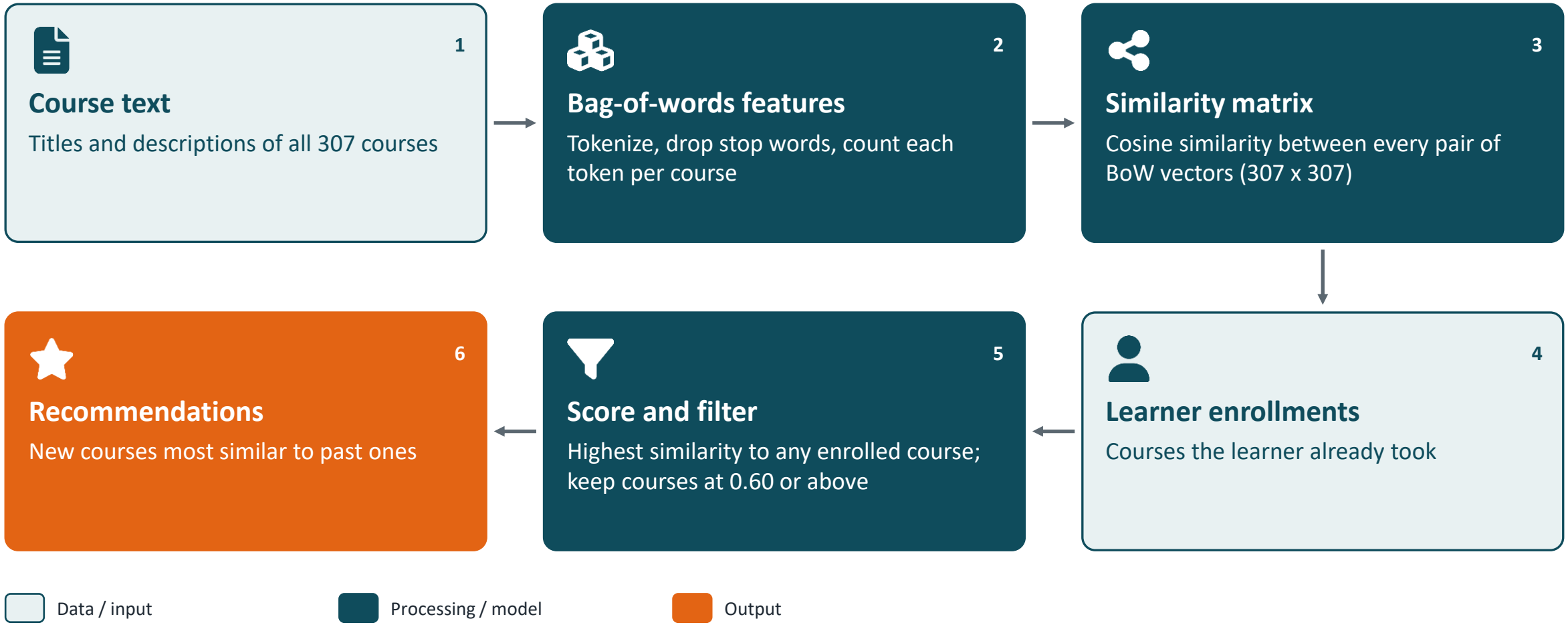
232
distinct courses ever recommended

Top 10 most frequently recommended courses (1,000 test learners)

#	Course	Times
1	Analyzing Big Data with SQL	234
2	Foundations for Big Data Analysis with SQL	234
3	Getting Started with the Data Apache Spark Makers Build	230
4	Analyzing Big Data in R using Apache Spark	218
5	Spark Overview for Scala Analytics	208
6	Cloud Computing Applications Part 2 Big Data and Applic…	190
7	Spark Fundamentals II	181
8	Big Data Capstone Project	152
9	Big Data Analysis Hive Spark SQL Dataframes and Graphfr…	152
10	Database Architecture Scale and NoSQL with Elasticsearch	152

Course similarity-based recommender

Flowchart: recommend courses whose content resembles what a learner already took



Course similarity-based: evaluation

Threshold 0.60 gives 11.4 courses per learner, and only 1.7% of learners get none

Hyperparameters

- **Similarity threshold:** 0.60, chosen from 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75 to give about 10 new courses per learner
- **Score:** highest cosine similarity to any course the learner took
- The catalog repeats some courses under different IDs, so near-duplicates appear in the top 10

11.4
new courses per learner (average)

10
new courses (median learner)

1.7%
of learners get no recommendation

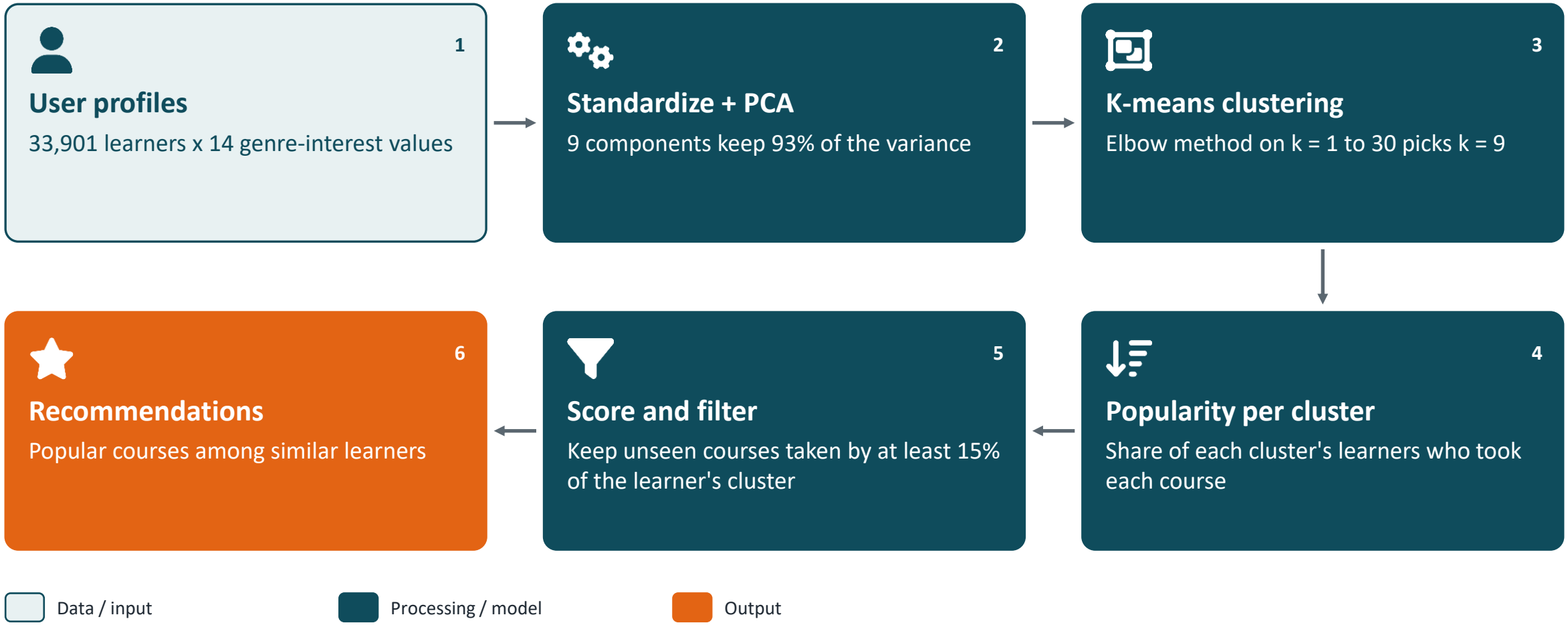
59
distinct courses ever recommended

Top 10 most frequently recommended courses (1,000 test learners)

#	Course	Times
1	Introduction to Data Science in Python	579
2	Introduction to Data Science in Python	579
3	Data Science with Open Data	562
4	A Crash Course in Data Science	555
5	Data Science Fundamentals for Data Analysts	555
6	Foundations for Big Data Analysis with SQL	551
7	Big Data Modeling and Management Systems	550
8	Introduction to Big Data	539
9	Fundamentals of Big Data	539
10	SQL Access for Hadoop	506

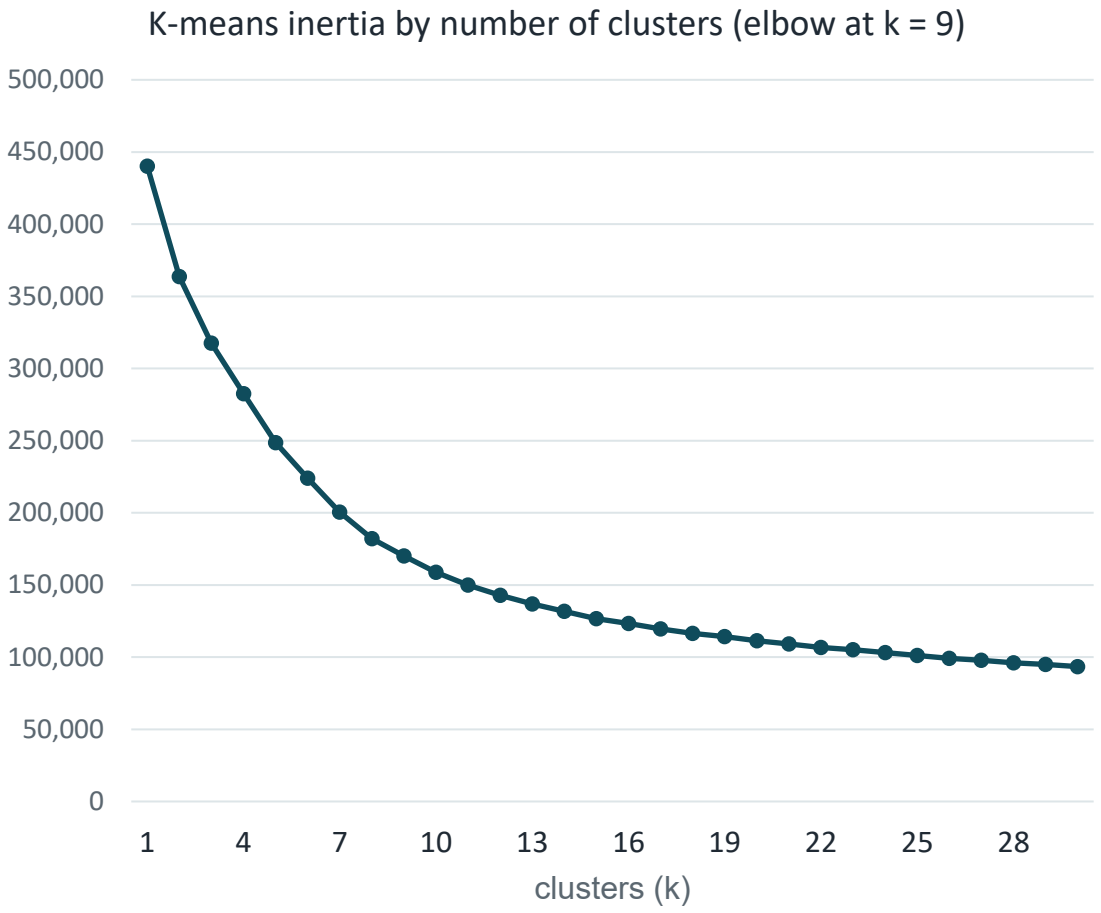
Clustering-based recommender

Flowchart: recommend what is popular among learners with similar interests



Clustering: learner segments

PCA keeps 9 of 14 components (93% of variance), and the elbow is at k = 9



Clusters, largest first

Cluster	Learners	Avg courses	Strongest interests
1	11,375	2.0	Light users (below average in every genre)
8	7,284	7.4	Python, Data science
6	4,587	8.1	Big data, Database
2	3,481	8.7	Containers, Cloud computing
5	3,128	8.6	Chatbot, Python
3	2,552	11.7	R, Data science
0	827	29.1	Data analysis, Database
7	647	11.7	Front-end dev, Back-end dev
4	20	16.0	Computer vision, Back-end dev

Clustering-based: evaluation

Threshold 15% gives 7.9 courses per learner, recommending popular courses within each segment

Hyperparameters

- **Clusters:** k = 9 (elbow), on 9 PCA components
- **Popularity threshold:** 15% of the cluster, chosen from 10%, 15%, 20%, 25%, 30%, 35%, 40%
- **Score:** share of the learner's cluster enrolled in the course

7.9
new courses per learner (average)

8
new courses (median learner)

8.1%
of learners get no recommendation

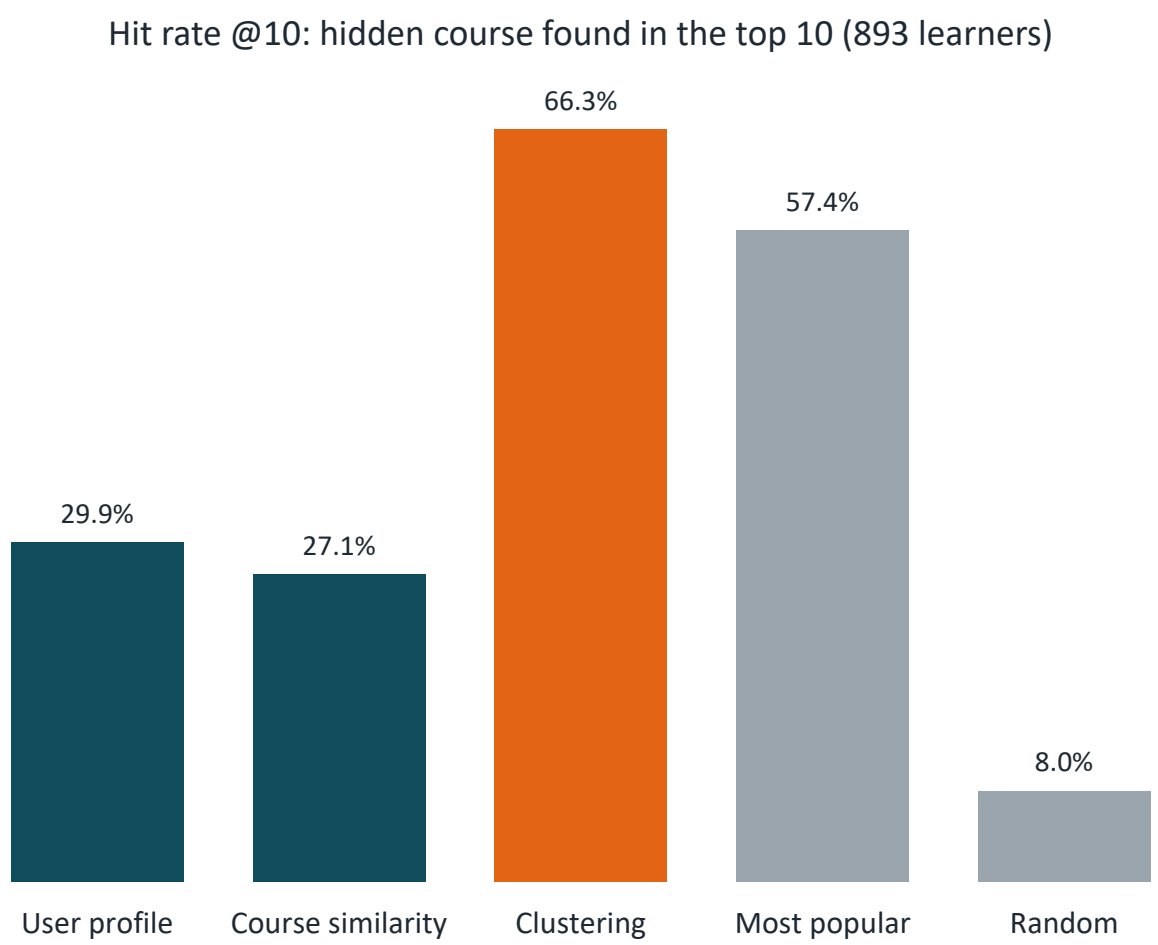
68
distinct courses ever recommended

Top 10 most frequently recommended courses (1,000 test learners)

#	Course	Times
1	Big Data 101	461
2	Introduction to Data Science	449
3	Hadoop 101	447
4	Data Science Methodology	425
5	Blockchain Essentials	424
6	Machine Learning with Python	373
7	SQL and Relational Databases 101	368
8	Statistics 101	338
9	Spark Fundamentals I	329
10	Deep Learning 101	321

Comparing content-based recommenders

Clustering recovers the most hidden enrollments: 66% vs 57% for most-popular



Method	Avg recs	No recs	Distinct	Hit @10
User profile	8.2	60.3%	232	29.9%
Course similarity	11.4	1.7%	59	27.1%
Clustering	7.9	8.1%	68	66.3%

- **Hold-out test:** hide one enrolled course per learner and check whether it appears in each method's top 10.
- **Learners' next courses are popular foundations**, so genre and text matching alone underperform even the most-popular list.
- **Clustering adds personalization to popularity** and beats it by 9 points.

03

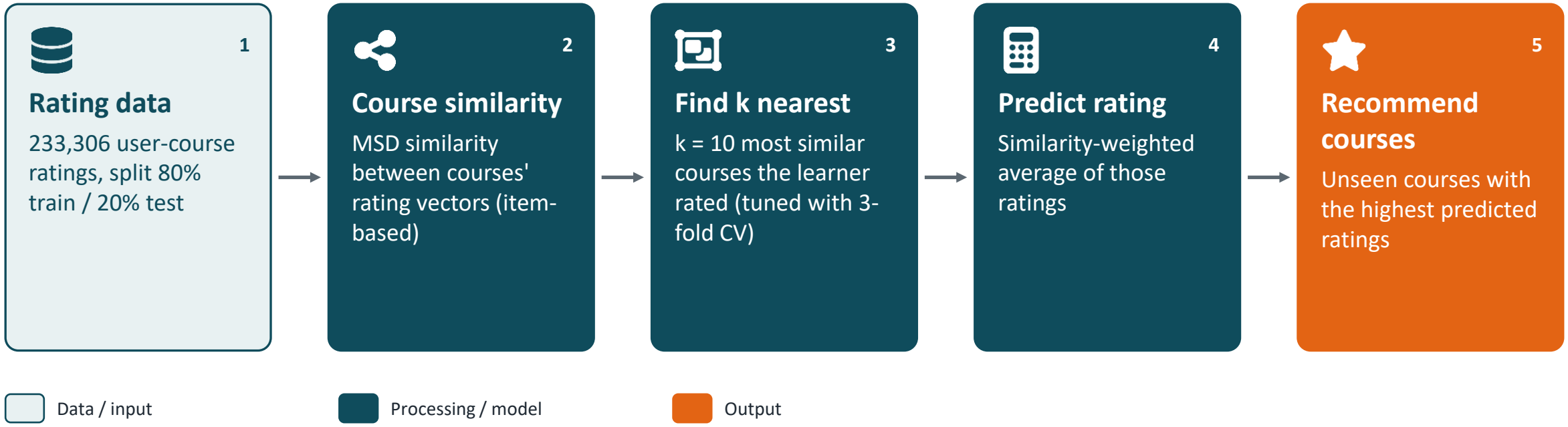
Collaborative-filtering Recommender Systems

Supervised learning from the ratings of all 33,901 learners



KNN-based recommender

Flowchart: predict a rating from the learner's ratings of the most similar courses



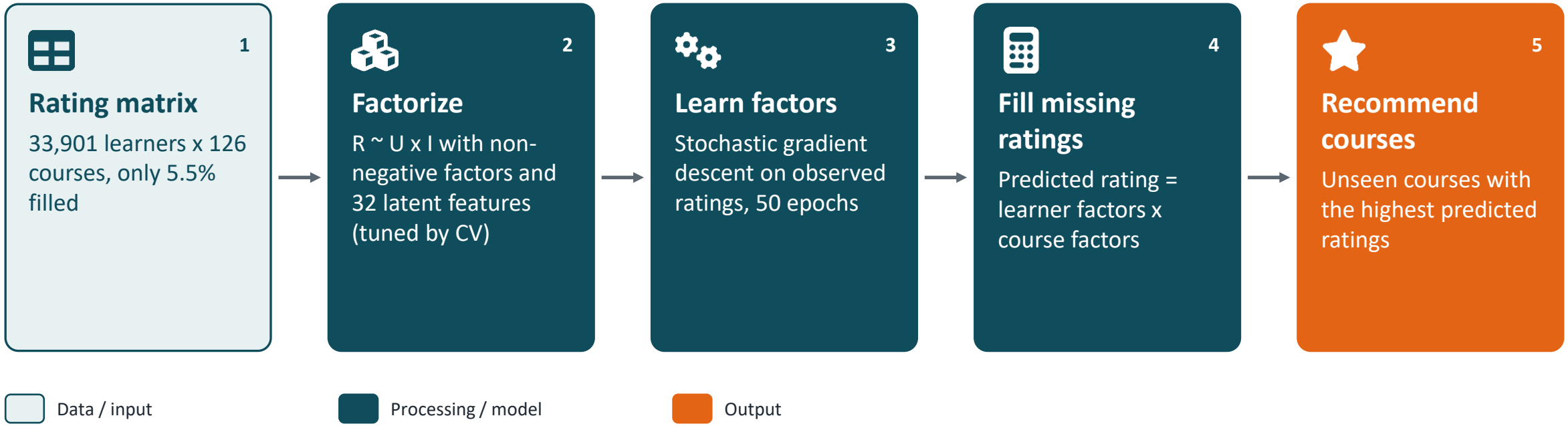
How a rating is predicted

predicted r(u, i) = Σ sim(i, j) · r(u, j) / Σ sim(i, j), over the k = 10 courses j most similar to course i that learner u rated

Item-based, because a user-based model would need a 33,901 x 33,901 similarity matrix

NMF-based recommender

Flowchart: factorize the sparse rating matrix into learner and course factors



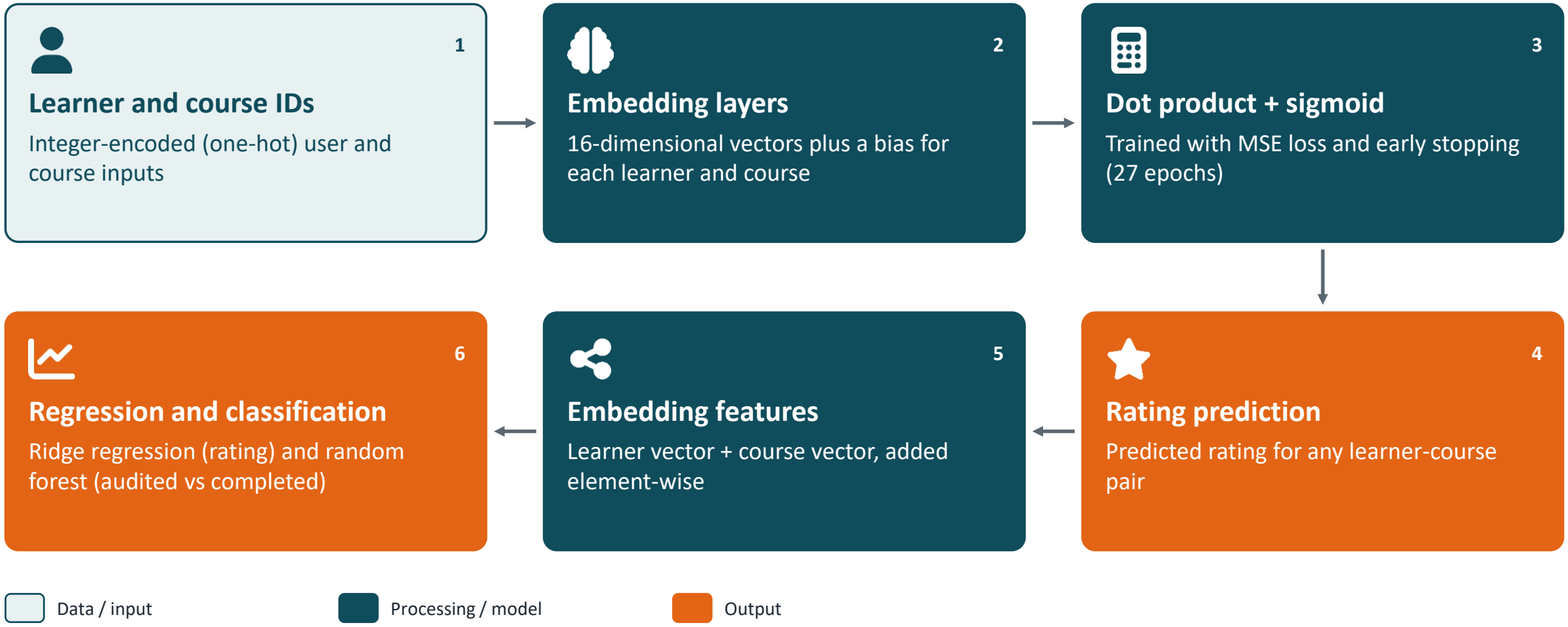
The factorization

R (33,901 x 126) ≈ U (33,901 x 32) × I (32 x 126), with every factor ≥ 0

predicted r(u, i) = learner u's factor vector · course i's factor vector

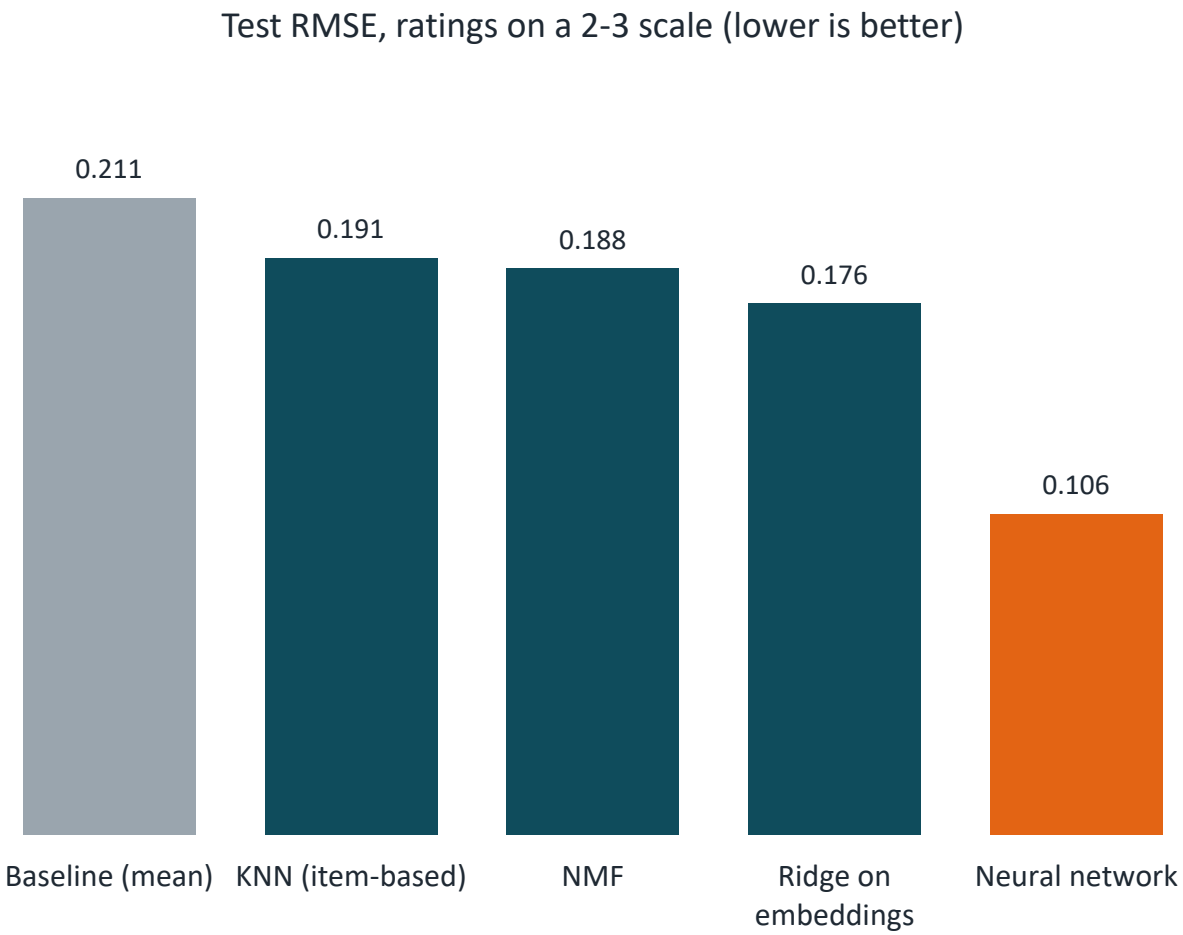
Neural network embedding recommender

Flowchart: learn embedding vectors for learners and courses, then reuse them as features



Comparing collaborative-filtering models

Neural network embeddings cut test RMSE by 50% versus always predicting the average rating



Classifying audited (2) vs completed (3)

Model	Accuracy	Balanced acc.	Audit recall
Majority class	95.3%	50.0%	0%
Logistic regression	82.5%	80.7%	79%
Random forest	99.3%	94.8%	90%

- **Embeddings capture learner habits:** the network's user and course vectors give the lowest RMSE (0.106 vs 0.211).
- **KNN and NMF give modest gains**, because almost every rating is 3.
- **Accuracy is misleading here:** always predicting "completed" scores 95% but never finds an audit. The random forest finds 90% of them.

Conclusions

Popularity among similar learners and learned embeddings carry the most signal

Enrollments are concentrated

The top 20 courses take 63% of enrollments, and 44% of learners took 5 or fewer courses.

Clustering is the best content-based method

It finds the hidden course in the top 10 for 66% of learners, ahead of most-popular (57%).

Embeddings win collaborative filtering

The neural network halves the baseline RMSE, and its embeddings let a random forest flag 90% of audits.

Next steps

Recommend top-N lists instead of thresholds, de-duplicate the catalog, and build a hybrid of cluster popularity and embeddings.

Appendix

Model settings and how to reproduce every number in this deck

Model	Hyperparameters	Value	Metric	Result
User profile	Score threshold	25	Avg recs / learner	8.2
Course similarity	Similarity threshold	0.60	Avg recs / learner	11.4
Clustering	k (PCA: 9 comp.)	9 / 15%	Avg recs / learner	7.9
KNN (item-based)	k, similarity	10, MSD	Test RMSE	0.1910
NMF	Factors, epochs	32, 50	Test RMSE	0.1875
Neural network	Embedding dim, epochs	16, 27	Test RMSE	0.1062
Ridge on embeddings	Alpha (CV)	0.1	Test RMSE	0.1760

- **Notebook:** course_recommender_capstone.ipynb runs every analysis in this deck end to end.
- **Data:** IBM ML321EN capstone datasets (ratings, course genres, course text, bag of words, user profiles, test learners).
- **Tools:** Python, pandas, scikit-learn, Surprise (KNN, NMF), Keras 3.15.1 (torch backend), wordcloud.
- **Reproducibility:** random seed 42; 80/20 train/test split for collaborative filtering.